A computational model of the interactions between formate and insulin metabolism

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Abstract

Formate, the circulating carrier of one-carbon units, supplies two carbons of every purine ring, and cellular purine nucleotides are sensed by mTORC1 — the regulator of the anabolic translation the pancreatic β-cell uses to make insulin. These links suggest an untested chain, formate $\rightarrow$ purine nucleotides $\rightarrow$ mTORC1 $\rightarrow$ insulin synthesis, connecting one-carbon nutrition to insulin production and diabetes risk. We formalise this chain as a computational kinetic model calibrated to independent islet data; the model itself — a quantitative substrate for reasoning about formate and insulin — is the central contribution. Systemically, raising formate increases the purine pool, mTOR activity and insulin synthesis, while loss of mitochondrial formate production cuts insulin synthesis by roughly half with a parallel fall in uric acid. Embedded in a whole-body glucose–insulin loop, this central action raises plasma insulin, distinguishing it from a peripheral effect on glucose uptake. An adipose compartment shows that obesity and an insulin-secretion defect drive blood formate in opposite directions — reconciling human cohorts that disagree in sign — while food formate content does not predict diabetes risk, because the β-cell is already formate-replete. A gut-microbiome compartment that returns urate carbon as formate closes a delayed positive-feedback urate cycle. The retina is distinct: it makes its own insulin behind the blood-retinal barrier, so retinal insulin tracks blood one-carbon rather than plasma insulin. Self-regulation makes the eye a homeostat; because excess retinal insulin is pathological, retinal health is an inverted-U in formate, and methanol optic toxicity emerges as an acute one-carbon overload. The model turns these maps into testable interventions: formate supplementation benefits only the formate-deficient diabetic, and in the eye, inhibiting local formate synthesis versus insulin synthesis is a paired, mechanism-discriminating prediction.

Author summary

Insulin is made by specialised pancreatic cells, and too little of it causes diabetes. I asked whether a small, common nutrient — formate, the body's carrier of one-carbon chemical groups — helps set how much insulin these cells can build. Formate supplies part of every purine, a building block of DNA and energy molecules, and cells read their purine levels to switch on the growth machinery (mTOR) that manufactures insulin. I built a computer model of this chain, tuned to published measurements from insulin-producing cells, and used it to explore what happens as formate rises and falls. The model reproduces puzzling human data — why blood formate can look high in some people with diabetes and low in others — and predicts who might benefit from formate supplements and who would not. It also speaks to the eye, which makes its own insulin: too little or too much formate harms the retina, and the model recasts methanol blindness as an insulin-excess overload rather than direct poisoning of the cell's respiratory chain. Most usefully, it proposes specific, testable experiments to confirm or refute each prediction.

Introduction

Insulin is the body's principal anabolic hormone, and the pancreatic β-cell that produces it is a specialised secretory factory that couples glucose sensing both to the release of stored insulin and to the synthesis of new hormone [1]. Proinsulin biosynthesis is among the most demanding acute anabolic tasks in mammalian physiology: a rise in glucose increases total β-cell protein synthesis roughly twofold but proinsulin synthesis ten- to twentyfold, almost entirely through translational control [2]. Such rapid, large-scale anabolic translation is the domain of mTOR complex 1, the master regulator that gates protein synthesis to nutrient and energy supply [3]; accordingly, mTORC1 signalling sustains β-cell mass, insulin content and glucose-stimulated secretion.

When insulin supply fails to meet demand, type 2 diabetes results. Obesity is the dominant driver: an expanded adipose mass releases fatty acids, cytokines and hormones that impose systemic insulin resistance and so raise the secretory demand on the β-cell [4]. Diabetes nevertheless emerges only when the β-cell can no longer compensate — β-cell dysfunction is the pivotal lesion — within a disease that is ultimately multi-organ, spanning islet, liver, muscle, fat, gut and brain [5,6]. Because insulin *production* sits at the heart of this failure, any nutritional input that modulates the β-cell's anabolic capacity is of potential relevance to diabetes risk.

One candidate input is formate, the circulating carrier of one-carbon units. Folate-mediated one-carbon metabolism supplies the carbons for nucleotide, amino-acid and methylation reactions and is highly compartmentalised: mitochondrial serine catabolism (SHMT2–MTHFD2–MTHFD1L) oxidises serine-derived one-carbon units to formate, which is exported to the cytosol and to the blood, and is complemented by dietary and gut-microbial sources [7,8]. Two carbons of every purine ring derive from formate, so formate availability constrains de novo purine synthesis [9]. In healthy adults, serum formate averages $\sim$26 μM (range 9–97 μM) and is set by its metabolic precursors (serine, methionine, choline) and by folate status rather than by smoking or alcohol [10], and circulating formate has been reported to stratify with cancer and diabetes [11].

These strands converge on a testable chain. Cellular purine nucleotide levels are directly sensed by mTORC1 [12], the very complex that drives insulin translation. If formate feeds the purine pool and purines raise mTORC1 activity, then formate could act as a nutritional modulator of insulin biosynthesis: **formate $\rightarrow$ purine nucleotides $\rightarrow$ mTORC1 $\rightarrow$ insulin peptide synthesis**. We proposed this chain informally [13]; it has not been tested directly. Here we build a computational model of it — the central contribution of this work — and use it to investigate, in turn, systemic insulin metabolism across formate states, retinal insulin metabolism across formate states, and the therapeutic implications that follow.

Results

A computational model of the formate – insulin chain

The central contribution is the model itself: a kinetic formalisation of the proposed formate $\rightarrow$ purine $\rightarrow$ mTOR $\rightarrow$ insulin chain, calibrated to independent islet data, that turns a verbal hypothesis into quantitative predictions we interrogate in the parts that follow.

A kinetic β-cell model coupling formate, purines, mTOR and insulin

The model (`BetaCellModel`) is a system of ordinary differential equations for 14 species in a single β-cell, with glucose as the external input and blood formate and mitochondrial formate production as levers. Five modules are coupled:

1. **Energy metabolism.** Glucose is committed by a glucokinase-like sensor (Hill, halfsaturation 8 mM) feeding glycolytic and oxidative ATP production; ATP, ADP and AMP are linked by adenylate kinase. The ATP/ADP ratio is both the anabolic fuel and, through a Hill trigger, the K-ATP/Ca^{2+} secretion signal.

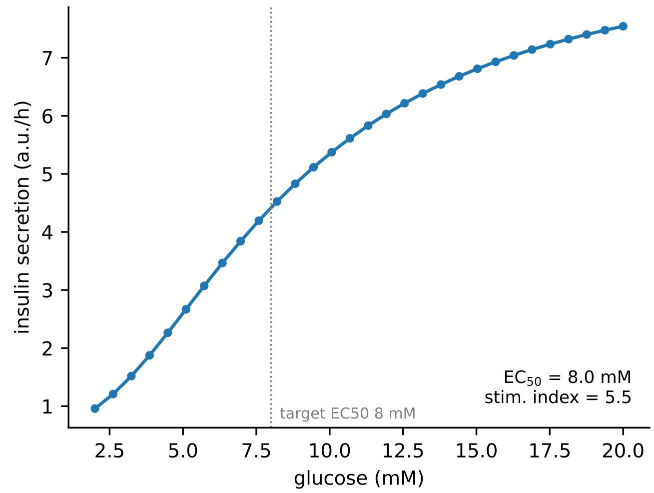

Fig 1. Calibration to GSIS. Steady-state insulin secretion versus glucose after fitting to islet-perifusion benchmarks; the fitted half-maximal glucose (EC_{50}) and stimulation index are shown.

2. **Purine synthesis.** The formate → 10-CHO-THF → PRPP → PPAT → GART → ATIC block of the Formate-NUDT5 model, including the optional AMP/PRPP-gated PPAT–NUDT5 metabolite glue, terminating in IMP.
3. **Purine degradation.** IMP, AMP and guanine nucleotides converge on hypoxanthine/xanthine and are oxidised to urate.
4. **Uric acid secretion.** Urate is exported from the cell.
5. **Insulin synthesis.** mTORC1 activity is a Hill function of the purine nucleotide pool (weighted to the guanine-nucleotide, de-novo-controlled pool) gated by the adenylate energy charge; it drives translation of pro-insulin, which matures and is secreted under the glucose trigger.

Every species has an explicit rate law (mass-action / Michaelis–Menten / Hill). The model is integrated in time to simulate transients and solved at steady state for dose-response analysis.

Calibration to independent islet datasets

Four modules were anchored to independent measurements, in dependency order. (1) *Energy*: the two energy parameters were fitted so the β-cell ATP/ADP ratio rises from **2.4 to 11.6** across 1→10 mM glucose [14], with the guanine-nucleotide pool rising in parallel [15] and physiological ATP (0.6→2.3 mM). (2) *IMP branch*: the adenylosuccinate-synthase/lyase rates were fitted so a low → high glucose step drops IMP ($\times$**0.25**) and raises adenylosuccinate/S-AMP ($\times$**3.4**), reproducing Gooding et al. [16]; this required an explicit, saturable S-AMP pool. (3) *mTOR → content*: the mTOR-independent fraction of insulin synthesis was fitted so complete mTORC1 loss (Raptor-KO, mTOR activity →0) lowers insulin content to **50%** of wild type [17]. (4) *GSIS*: the glucose half-saturation and basal drive of the insulin gene were fitted to the concentration-dependent (second-phase) perifusion response — half-maximal secretion near 8 mM (human islets 7.9 mM) and a stimulation index ∼5–6 [18] — giving $\mathbf{EC_{50}}$ = 8.0 **mM, stimulation index** = 5.5 (Fig 1). The transient first phase (peak ∼1 min) is below the model's time resolution and was not fitted. Anchoring the granule pool to the measured β-cell insulin content (∼20 pg/cell) puts secretion in per-cell units without changing any calibrated ratio.

Systemic insulin metabolism across formate states

We first ask how systemic insulin and glucose respond as formate is varied — through dietary supply, mitochondrial production, and the disease states that move blood formate — from the isolated β-cell out to the whole body.

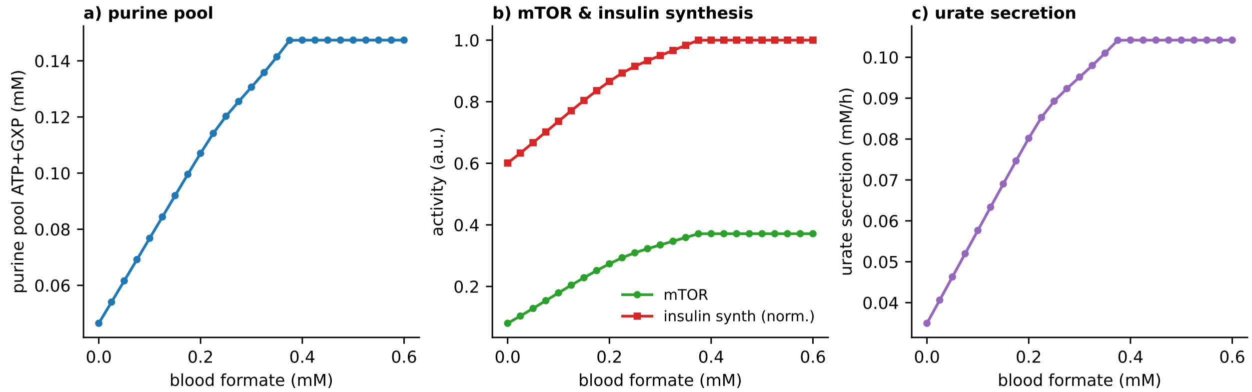

Fig 2. Formate stimulates insulin synthesis. Dietary-formate dose-response at 11 mM glucose on a low-endogenous-formate background. (a) purine pool; (b) mTOR activity and normalised insulin synthesis; (c) urate secretion.

Formate stimulates insulin synthesis

At stimulatory glucose (11 mM) on a formate-limited background, raising blood formate increased the purine nucleotide pool, mTOR activity ($0.08 \rightarrow 0.37$) and the insulin-synthesis rate by $\sim$66 %, saturating thereafter (Fig 2). Uric acid secretion rose in parallel, from 0.035 to 0.104 mM/h. The response is steep across the physiological blood-formate range (≤ 0.3 mM), supporting the hypothesis that formate availability sets an anabolic ceiling on insulin production.

Mitochondrial formate deficiency impairs insulin (formate theory of diabetes)

We simulated formate deficiency by suppressing the endogenous mitochondrial formate source (the MTHFD1L pathway). Lowering mitochondrial formate production from normal to zero, at fixed 8 mM glucose, reduced intracellular formate, collapsed mTOR activity ($0.35 \rightarrow 0.03$) and cut insulin synthesis by $\sim$45 % (Fig 3) — the fall bounded by the mTOR-independent synthesis floor calibrated to Ni *et al.* Urate secretion fell in parallel ($0.10 \rightarrow 0.02$ mM/h). The model thus predicts that a defect in endogenous formate production — genetic (MTHFD1L variants), mitochondrial, or environmental (antibiotic depletion of formate-producing microbiota) — produces a coupled low-insulin, low-urate state, consistent with the reduced circulating formate reported in diabetes and its seesaw with glucose [11]. Providing dietary formate rescued insulin synthesis on the deficient background (Fig 2), the in-silico analog of sodium-formate supplementation.

In a glucose-tolerance transient, a formate-replete cell entered the glucose bolus with higher basal mTOR and a fuller granule pool and mounted a roughly two-fold larger integrated insulin response than a formate-deficient cell (Fig 4), linking chronic formate status to acute glucose tolerance.

Sensitivity analysis isolates the controlling parameters

Varying each assumed parameter two-fold (`data/sensitivity.csv`), we tracked two outputs: the GSIS stimulation index (SI) and the formate sensitivity of insulin synthesis (FS, the ratio of insulin synthesis at replete vs deficient formate). Both outputs were insensitive (relative change < 1 %) to every insulin-secretion kinetic constant (maturation, secretion rate, secretion trigger, ATP cost). SI was controlled by the energy/mTOR parameters (basal fuel floor, purine-sensing Hill coefficient and half-saturation), and FS was controlled specifically by the purine-synthesis and mTOR-sensing parameters (purine turnover scale, mitochondrial formate capacity, the IMP-branch and AMP-degradation rates, and the purine-sensing Hill coefficient). The formate $\rightarrow$ insulin prediction therefore rests on the identifiable purine/mTOR module and not on the loosely-constrained secretion kinetics.

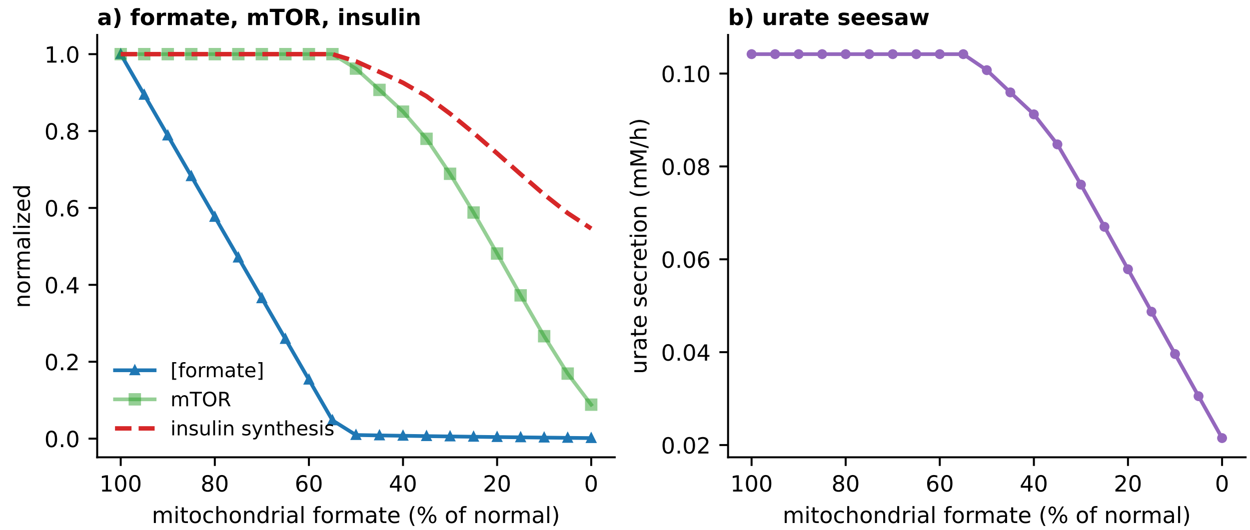

Fig 3. Mitochondrial-formate deficiency impairs insulin. Suppressing the endogenous (MTHFD1L) formate source at 8 mM glucose. (a) intracellular formate, mTOR and insulin synthesis (normalised); (b) urate secretion falls with formate (the formate–urate seesaw). x-axis inverted: normal on the left, deficient on the right.

A whole-body loop discriminates central from peripheral formate action

The central prediction — formate improves glucose tolerance by raising β-cell insulin synthesis — must be separated from the alternative that formate (via methylamine/AOC3) raises *insulin-independent* peripheral glucose uptake [19]. We embedded the β-cell as the insulin source of a Bergman-type minimal model of blood glucose, plasma insulin and remote insulin action (`WholeBody`). In this framework the peripheral mechanism is precisely an increase in glucose effectiveness (S_g), the insulin-independent disposal term. We simulated a glucose-tolerance test under formate acting centrally (through the mechanistic β-cell), peripherally (raising S_g), or both, tuning the peripheral effect so that it lowered the glucose excursion by the same amount as the central one (Fig 5).

Both mechanisms lowered the glucose area-under-the-curve (central $-21\,\%$, peripheral $-14\,\%$), but they were cleanly separated by plasma insulin: the central mechanism *raised* the insulin AUC by $20\,\%$, whereas the peripheral mechanism left it essentially unchanged ($-2\,\%$; slightly lower, because the reduced glucose excursion drives less secretion). The model therefore yields a concrete discriminating experiment: **during a formate-supplemented glucose-tolerance test, measure plasma insulin** — a rise indicates the central (β-cell synthesis) mechanism of *The Spice of Life* [13], whereas unchanged or lower insulin at improved glucose tolerance indicates the peripheral mechanism of Carpéne *et al.* The two are not mutually exclusive; the "both" scenario shows their glucose effects add while the insulin signature follows the central arm.

Reconciling the formate–diabetes paradox

A large prospective cohort (Takase et al. [20]; TMM CommCohort, $n = 12{,}461$, 354 incident cases) reports circulating formic acid *positively* associated with incident type 2 diabetes (relative risk 1.45, 95% CI 1.11–1.91 per 1-SD), the *opposite* of the low-formate reading of the deficiency route. We asked whether the model can nonetheless reproduce this. The model admits two distinct routes to diabetes: (A) formate *deficiency*, which lowers insulin and predicts *low* formate; and (B) a primary insulin- *secretion* defect — the genetically driven diabetes of the cohort — which is independent of formate. Because insulin is the anabolic hormone that drives de novo purine synthesis, and purine synthesis is the principal formate *sink*, route (B) predicts that impaired insulin lets formate *overflow* upward.

We tested this by promoting blood formate to a dynamic pool — $\dot{F}_b = \text{intake} + \text{endogenous} - U(\text{insulin}, F_b) - k_{\text{ren}} F_b$, with the consumption U scaling with plasma insulin —

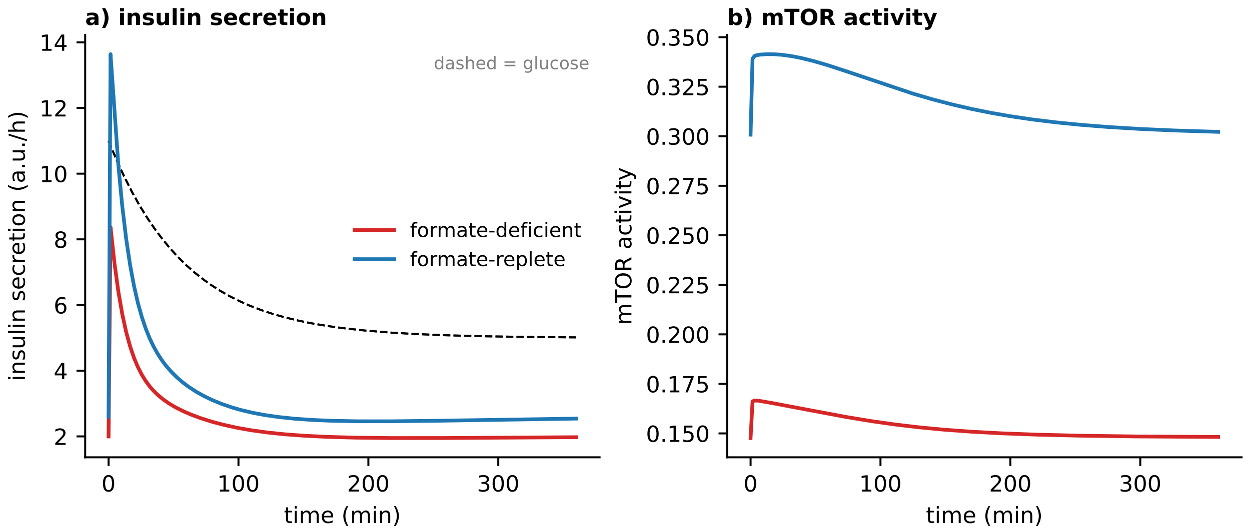

Fig 4. Formate priming and glucose tolerance. Glucose-tolerance transient (dashed: glucose bolus) for a formate-replete versus a formate-deficient cell, each pre-equilibrated to its basal steady state. (a) insulin secretion; (b) mTOR activity.

on top of the whole-body loop, and imposing a graded secretion defect (`FormateDiabetes`). As secretion is impaired, blood glucose rises (4.8→7.8 mM) and blood formate rises ×**1.5–1.6 at every dietary serine/formate intake** (0.8→1.3, 1.6→2.6 and 4.4→6.4 mg/L for low, normal and high intake; Fig 6), yielding a positive formate–glucose association from reduced formate *utilisation* rather than causation. This is consistent with the cohort's own internal evidence: formate did not differ by genetic risk (11.3 vs 11.2, $P = 0.77$) and was not a mediator of the polygenic-risk → diabetes effect, i.e. it behaves as a downstream consequence marker, exactly as route (B) predicts. The model therefore produces *either* low formate (deficiency-route diabetes [11]) *or* high formate (secretion-defect diabetes [20]), and the two are told apart by the accompanying insulin: overflow occurs with *low* insulin and high glucose.

Obesity and diabetes move blood formate in opposite directions

The two cohorts that bracket the human evidence disagree in sign: severe obesity carries *low* formate [11] whereas incident diabetes in a largely non-obese population carries *high* formate [20]. We asked whether a third compartment — adipose tissue — resolves this. We built a metabolic model of the adipocyte (`AdipocyteModel`) reusing the same energy and one-carbon/purine machinery as the β-cell but with insulin-dependent glucose uptake and lipid storage in place of insulin synthesis. Adipose plays two formate-relevant roles: it is a one-carbon *sink* (blood formate → de novo purine) and, through xanthine oxidoreductase, a uric-acid *source* whose activity and mass both rise in obesity [21]. Coupling the β-cell, the adipocyte and shared blood pools (glucose, insulin, formate, urate; `CoupledModel`), we crossed obesity (adipose mass ×3, XOR ×2, insulin resistance) with diabetes (secretion defect).

The two axes move blood formate oppositely (Fig 7). Obesity alone *lowers* blood formate (−14 %) — the expanded adipose sink consumes it — while *raising* blood uric acid (+4 %), matching Pietzke and Tsushima. Diabetes alone *raises* blood formate (+21 %) by the secretion-defect overflow of the previous section. In the combined obese-diabetic state the two nearly cancel (+5 %), and uric acid is highest (+5 %). The model thus predicts that the sign of the formate–diabetes association depends on how much obesity accompanies the diabetes: an obesity-dominated cohort sees low formate, a secretion-defect-dominated (leaner) cohort sees high formate — reconciling the two datasets within one mechanism, and simultaneously reproducing the obesity–hyperuricaemia link.

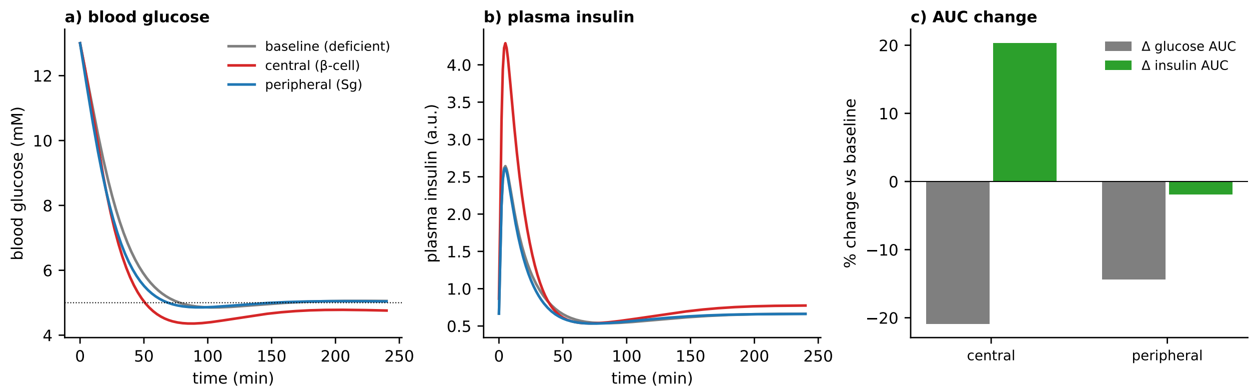

Fig 5. Plasma insulin discriminates the site of formate action. Whole-body glucose-tolerance test with formate acting centrally (β-cell insulin synthesis) or peripherally (glucose effectiveness S_g), matched to the same glucose lowering. (a) blood glucose (central $\approx$ peripheral); (b) plasma insulin (separates them); (c) change in glucose and insulin AUC versus baseline — same glucose, opposite insulin.

The gut microbiome closes a urate cycle: a delayed positive feedback

Urate is the end product of purine catabolism and is usually treated as a metabolic dead end, cleared renally and intestinally. But the gut microbiome degrades urate anaerobically, and this bacterial purine metabolism is now recognised as a major contributor to host purine homeostasis [22]. The *urate cycle* hypothesis [23] proposes that this closes a loop: of the five urate carbons the microbiome returns three — the two glycine-derived carbons as acetate (feeding lipogenesis) and one of the two purine formate carbons as formate itself — so that formate $\rightarrow$ de novo purine $\rightarrow$ urate $\rightarrow$ (gut) $\rightarrow$ formate becomes a cycle, with the co-produced acetate a candidate driver of the obese metabolic signature.

We added a gut-microbiome compartment (`GutUrateCycle`): blood urate is taken up by the gut route (diverting it from renal loss) and returned to the circulation as formate after a processing/transit delay, modelled as a linear chain of compartments (mean lag τ_{gut}). Three features emerge (Fig 8). First, the return *recovers urate carbon as formate*: switching the cycle on raises blood formate (lean $\sim$1.5 $\rightarrow$ 2.3 mg/L, obese $\sim$1.25 $\rightarrow$ 2.0 mg/L) while lowering circulating urate, now partly cleared through the microbiome rather than accumulating (Fig 8b). Second, it is a *bounded* positive feedback: because only one of the two purine formate carbons is recycled, the loop gain is below one, so blood formate rises smoothly and *converges* as the gut recycling strengthens — self-reinforcing but stable, not runaway (Fig 8c). Third, the gut delay makes it a *lagged* feedback: a purine/urate load (a purine-rich meal) clears from the blood within a few hours but returns as a secondary blood-formate rise $\sim\tau_{gut}$ ($\sim$12 h) later (Fig 8d) — a signature the closed loop predicts and against which a labelled purine bolus could be timed.

This places obesity inside the loop: the adipose xanthine-oxidoreductase induction that raises urate in obesity (above) supplies the cycle with more substrate, while the co-produced acetate feeds fatty-acid and cholesterol synthesis — the hypothesised route from urate to weight gain [23] — which nominates the gut urate step, rather than host xanthine oxidase, as a more specific therapeutic target. (The model captures the formate-recycling arm quantitatively; the acetate $\rightarrow$ lipogenesis arm is represented only as a noted output, and the delay, yield and gut-uptake rate are assumed, so the robust content is the qualitative behaviour: carbon recovery, bounded feedback, and delayed return.)

Diet: food formate does not predict systemic glucose

The dietary sources of formate differ enormously per serving. Whole foods dominate [13]: $\sim$1000 mg from 1 g creatine, 700 mg from 200 g meat, 70–600 mg per litre of fruit juice. Caffeine- and aspartame-containing drinks and medications contribute far less; for these we use the formate-generation potential tabulated by Pietzke et al. [8] (Table 2) — the formaldehyde released by demethylation of caffeine and aspartame, taken to full conversion to formate — which gives $\sim$43 mg per diet cola, $\sim$37 mg per diet soda, $\sim$31 mg per cup of brewed coffee, $\sim$8 mg per regular cola, and up to $\sim$47 mg per caffeine pill (Fig 9). (The commonly cited

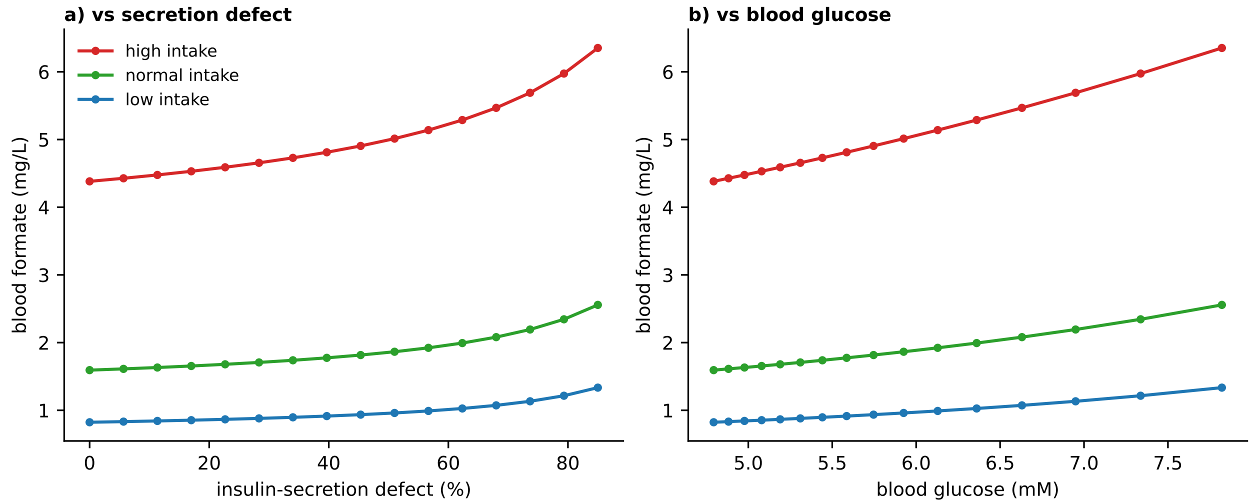

Fig 6. A secretion defect makes blood formate overflow. Blood formate versus the insulin-secretion defect (a) and versus blood glucose (b), at low/normal/high dietary serine–formate intake. Impaired secretion raises both glucose and formate at every intake, reproducing the positive formate–diabetes association of Takase et al. [20] as reduced insulin-driven consumption, not causation.

"80 mg per diet cola" is the aspartame *mass*, not the formate it yields.) If dietary formate set glucose control, these should rank foods by benefit. We tested this against a meta-analysis of the glucose/diabetes literature for each item and against the model.

The ranking fails. Pooling the best available estimates for incident type 2 diabetes per serving — coffee RR 0.91 (0.89–0.94 [24]), sugar-sweetened cola RR 1.13 and 100% fruit juice RR 1.07 and artificially sweetened (diet) cola RR 1.08 [25], red meat HR 1.10 per 100 g [26], and creatine glycaemically neutral in randomized trials (fasting-glucose SMD +0.05, HOMA-IR SMD −0.38, both non-significant [27]) — the correlation between a food's formate content and its diabetes effect is essentially zero ($r = -0.03$; Fig 10). Coffee, near the bottom of the formate table, is the only protective item; creatine and meat, at the top, are neutral and harmful. Each food's glycaemic effect is carried by its dominant components — sugar, saturated fat, chlorogenic acids — not its formate.

The model predicts exactly this null. Because the pancreatic β-cell is already formate-replete, adding dietary formate raises blood formate ($\sim$3-fold across the range of foods; Fig 9) but leaves plasma insulin and fasting glucose flat — unchanged across every food, and so not plotted; systemic glucose control is therefore governed by what else the food carries, not by its formate. Consistent with this, the two foods with any glycaemic benefit act by *insulin-independent* routes — creatine through GLUT4 translocation (with exercise), coffee through non-caffeine components (caffeinated and decaffeinated coffee are equally protective) — not through β-cell insulin. The one place dietary formate does register in the model is not systemic at all but ocular — a prediction we take up with the retinal model (below).

Retinal insulin metabolism across formate states

The retina is a special case. It makes its own insulin behind the blood-retinal barrier, so retinal insulin tracks blood one-carbon rather than plasma insulin. We give it a dedicated compartment and follow its insulin from deficiency, through a physiological optimum, to the toxic excess of methanol.

The eye: a self-regulating one-carbon compartment

The retina makes its own insulin: the retinal pigment epithelium (RPE) produces and secretes insulin locally, stimulated by photoreceptor phagocytosis, to support cone survival and regeneration, and it does so even under systemic insulin deficiency or resistance [28]. The blood-retinal barrier keeps *circulating* insulin out of the eye, so the retina cannot borrow plasma insulin — but small molecules, including glucose and one-carbon

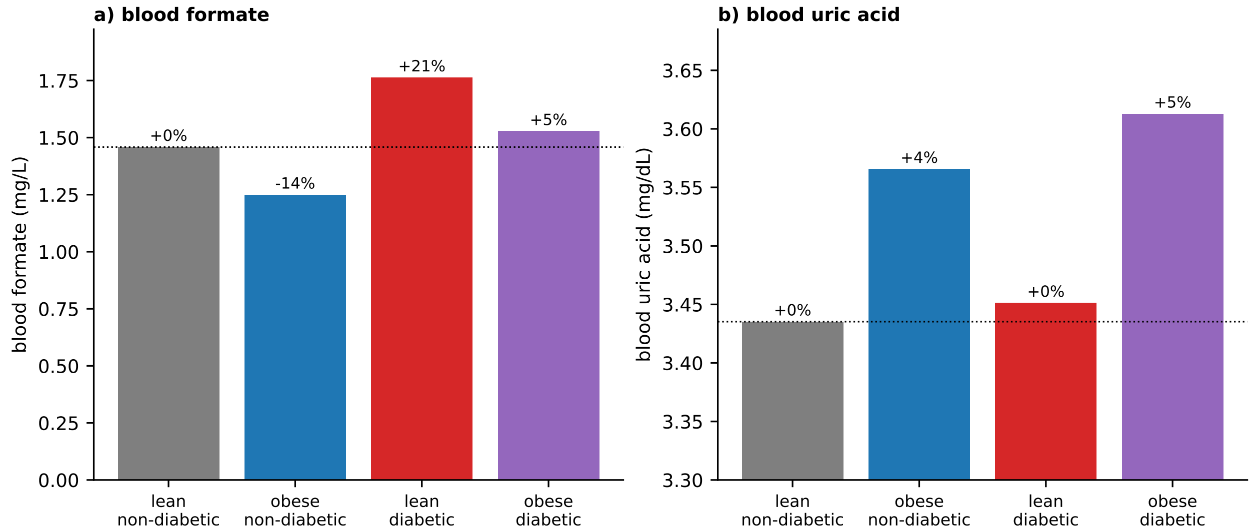

Fig 7. Obesity and diabetes push blood formate opposite ways. Three-compartment model (pancreas + adipose + blood). (a) blood formate falls with obesity (adipose one-carbon sink) and rises with diabetes (secretion-defect overflow); the obese-diabetic state is intermediate. (b) blood uric acid rises with obesity via adipose XOR [21]. Percentages are relative to the lean non-diabetic state.

units (serine/formate), do cross. We therefore added an eye compartment (`EyeModel`): a retinal insulin-producing cell running the same formate → purine → mTOR → insulin machinery as the β-cell, driven by blood glucose and blood formate, with its insulin kept strictly local and a low endogenous mitochondrial formate. Local insulin drives a retinal maintenance/regeneration index. Supplied only from the blood, eye insulin rises monotonically with circulating formate (Fig 12, blue).

Does the eye also make its own formate, co-regulated with the insulin it produces? Two lines of evidence say partly yes. First, single-cell expression (Human Protein Atlas; Fig 11): the insulin-producing RPE expresses the one-carbon *entry* enzymes SHMT2 and MTHFD2 at modest levels but is *depleted* for the formate-*releasing* enzyme MTHFD1L, which instead peaks in the adjacent cone photoreceptors (the eye's highest, $\sim$46× the RPE) — the very outer segments the RPE phagocytoses to trigger its insulin. Second, the mechanism: mTORC1 → ATF4 is the master inducer of SHMT2/MTHFD2 [29], and in islets the same ATF4 program that raises insulin content co-induces this serine-linked one-carbon pathway [30]; phagocytosing RPE activates ATF4 as its top upstream regulator. We could not find a dataset that directly measures these enzymes rising with *Ins2* in the RPE, so the co-induction is, for now, strong inference rather than eye-level measurement.

We encoded it as a feed-forward: the eye cell's local formate production is induced by its own mTOR/insulin-synthesis signal (lumping the RPE's ATF4-driven one-carbon entry with the cone-paracrine, MTHFD1L-rich formate supply). Left unregulated this runs away — eye insulin is pinned at its formate-saturated maximum ($\sim$2.62 a.u., well above the retinal optimum of $\sim$1.98; Fig 12, red) — which is biologically implausible: local retinal insulin is homeostatically controlled. Two mechanisms do so, both documented in the RPE. Production is not constitutive but *signal-gated* — switched on by MerTK-dependent phagocytosis of photoreceptor outer segments, a circadian pulse amplified by starvation [28]. And insulin/IGF signalling is capped by the mTORC1 → S6K1 → IRS-1 negative-feedback loop, shown directly in RPE cells [31]; indeed the proliferative retinopathy of chronic ocular IGF-1 excess [32, 33] reads as loss of exactly this cap. Adding the S6K1 → IRS-1 feedback tames the feed-forward: across the physiological range of blood formate the regulated eye holds local insulin near the retinal optimum ($\sim$1.9–2.0; Fig 12, green) — *lifted* toward the optimum at the low circulating formate of obesity or metformin, where the blood-dependent eye is deficient, and overridden into excess only when *external* formate climbs above the physiological range.

Because a positive feed-forward can in principle produce bistability (a switch with hysteresis), we characterized the regulated eye before using it. Basin tests (integrating from many initial conditions at each

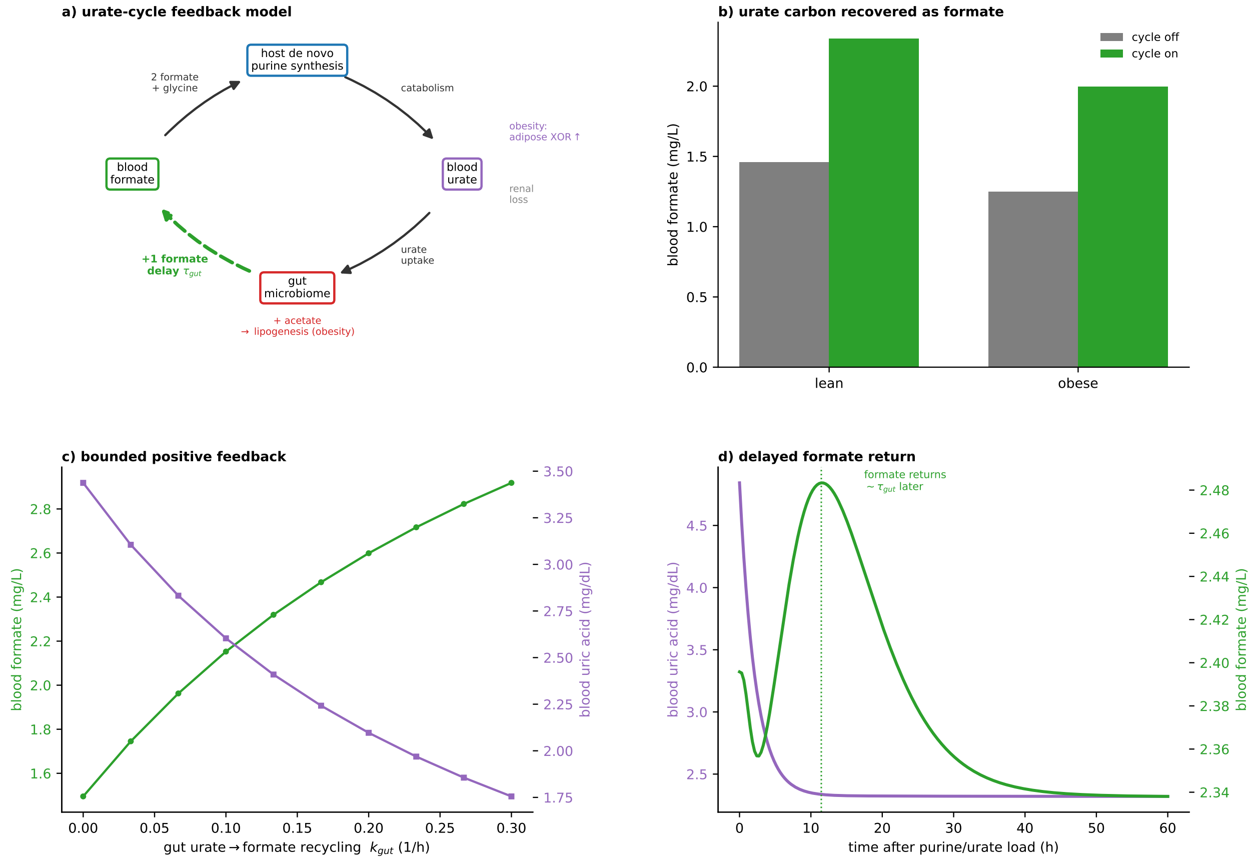

Fig 8. The gut microbiome closes a urate cycle: a delayed positive feedback. A gut-microbiome compartment degrades blood urate and returns formate after a delay τ_{gut}. (a) Schematic of the model: host purine synthesis consumes two formate (plus glycine) per urate; the gut microbiome returns one of them as formate after the delay τ_{gut} (green, closing the loop) and co-produces acetate (→ lipogenesis). (b) Steady-state blood formate with the cycle off versus on, lean and obese — urate carbon is recovered as formate. (c) As the gut urate → formate recycling k_{gut} strengthens, blood formate rises and blood urate falls, both converging: a bounded positive feedback (loop gain < 1). (d) A purine/urate load clears from the blood within hours but returns as a blood-formate rise $\sim\tau_{gut}$ ($\sim$12 h) later.

condition) and a self-consistency/bifurcation analysis agree (Fig A in S1 Appendix): the response is *monostable* — a single stable steady state at every blood formate and glucose, with no hysteresis, and no bistability even at five-fold the reference feed-forward gain (the S6K1 cap prevents it). The regulated eye is therefore a smooth homeostat, not a switch; were the retina's insulin pulse switch-like (it is circadian), that behaviour would come from the phagocytosis signal-gate, not from this metabolic loop.

Across the clinical conditions the prediction is a clean dissociation (Fig 13). The regulated eye insulin sits near the retinal optimum but is decoupled from *plasma* insulin: it *anti*-correlates with plasma insulin ($r = -0.90$) and tracks blood one-carbon and local glucose ($r = +0.89$ with blood formate). The lean healthy and obese non-diabetic states, which carry the *highest* plasma insulin, sit just *below* the optimum (mild deficiency); the lean secretion-defect diabetic — lowest plasma insulin, but highest blood formate and hyperglycaemic — is pushed *above* the optimum into mild retinal-insulin excess. Retinal insulin status is thus governed by blood one-carbon and local glucose, not by systemic insulin: states that lower blood formate (obesity's adipose sink; metformin's SHMT2 inhibition) pull the retina toward the deficient side even as systemic glucose improves, while serine/formate raise it (a candidate mechanism relevant to diabetic retinopathy, and consistent with the ocular tolerability of formate supplementation [34]).

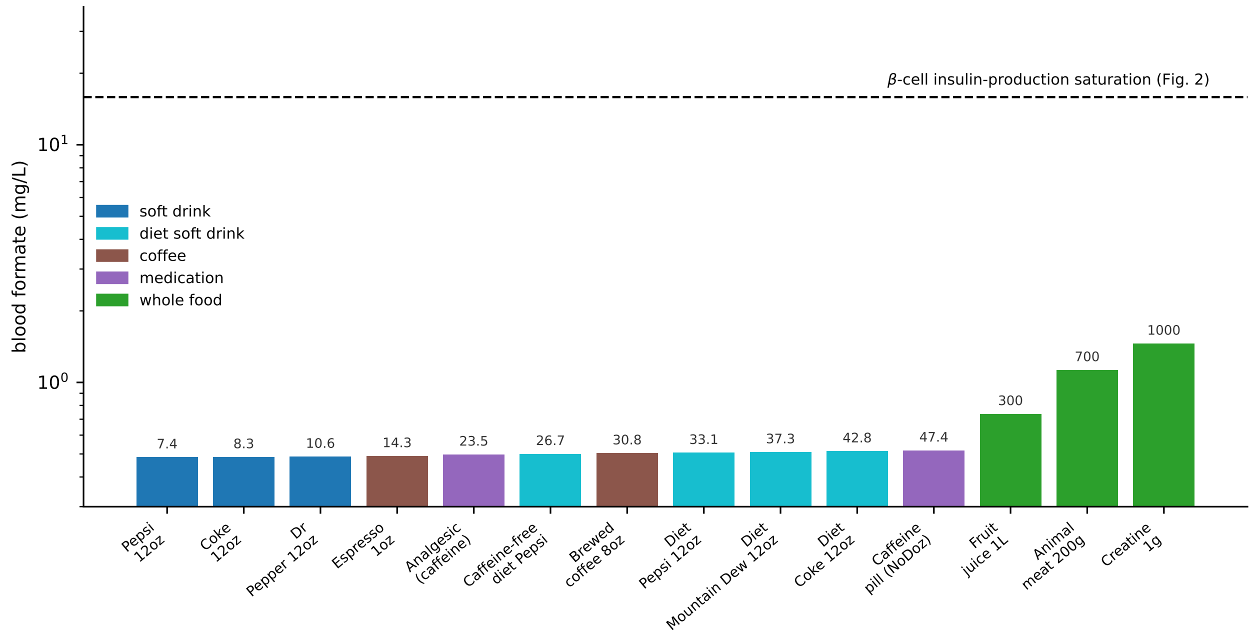

Fig 9. Blood formate delivered by each food/beverage. Model-predicted blood formate for the caffeine/aspartame drinks and medications of [8] Table 2 (coloured by category, formate per serving annotated) plus the whole-food sources; all the beverages and pills deliver near-baseline blood formate, spanning ~3-fold only once the high-content whole foods (fruit juice, meat, creatine) are included. The dashed line marks the blood-formate level that saturates β-cell insulin production (Fig 2, ~0.3 mM); every food lies more than an order of magnitude below it (note the log axis), so dietary formate cannot drive systemic insulin.

Retinal health: an optimal insulin window

High formate does something specific in the eye: it drives local insulin toward its saturated maximum (Fig 12). An *excess* of retinal insulin signalling is itself pathological, on firm evidence. Chronic ocular over-expression of IGF-1/insulin drives intraocular VEGF and a progression to proliferative, neovascular retinopathy even without hyperglycaemia [32]; sustained retinal IGF-I causes gliosis, oxidative stress and neurodegeneration despite being neuroprotective at normal levels [33]; and the "early worsening" of retinopathy on intensive insulin is attributed to insulin acting with VEGF to drive vascular proliferation. Retinal insulin is thus beneficial in a window and harmful outside it.

We encoded this as a trophic maintenance arm that saturates with local insulin, minus a pathology arm (VEGF/gliosis/neovascularization) with a higher threshold, so net retinal health is an inverted-U in local insulin (Fig 14). Because eye insulin rises with blood formate, retinal health peaks at a physiological formate (~5–10 mg/L) and falls off on both sides: *deficiency* at low formate (obesity's sink, metformin) and *insulin excess* at high formate (over-supplementation, and the methanol range), where net health drops to ~0.24 against a peak of ~0.42. This inverted-U is the "optimum" that the homeostat of the previous section targets, and it unifies the eye story: the retina, like the rest of the body in this model, needs formate — and the local insulin it sets — within an optimal window, with disease at either end. It notably requires *no* complex-IV inhibition (a mechanism that rests on sparse, largely in-vitro evidence; see Discussion), resting instead on the well-supported biology of retinal insulin/IGF excess. It also predicts a mechanism for diabetic retinopathy orthogonal to hyperglycaemia: a formate/eye-insulin excess (as in the lean secretion-defect overflow) or deficiency (as in obesity) could damage the retina at normal plasma insulin.

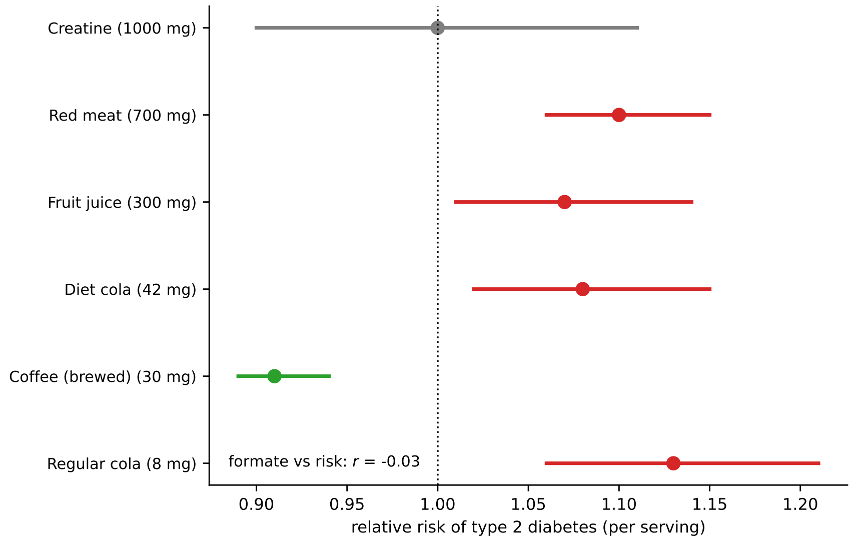

Fig 10. Food formate does not predict the diabetes effect. Meta-analytic relative risk of type 2 diabetes per serving for the six items with a per-serving estimate (beverage formate contents from [8] Table 2), ordered by formate content (mg per serving in parentheses); the formate ranking is uncorrelated with the effect ($r = -0.03$). Coffee (low formate) is protective while creatine and meat (high formate) are neutral and harmful.

Methanol as an acute one-carbon overload

The same molecule that supports the eye at physiological levels is the toxin of methanol poisoning at high ones: methanol is oxidised to formate, which accumulates to the symptomatic (> 50 mg/L) and lethal (> 500 mg/L) levels of intoxication and produces the classic visual loss. In this model that ocular damage is *insulin excess*, not inhibition of mitochondrial complex IV: we leave the respiratory chain uninhibited (the complex-IV route rests on sparse, largely in-vitro evidence; see Discussion) and let rising formate act only by driving local insulin past the retinal optimum into the pathological excess zone (the well-evidenced retinal-health window, Fig 14). We swept blood formate from physiological to the methanol range for the blood-dependent eye and for the S6K1-regulated autonomous eye (Fig 15).

Regulation splits the response into two regimes. Across the *physiological* range the regulated autonomous eye holds insulin at the optimum, so its net retinal health sits at the peak ($\sim$0.42) even at low blood formate, whereas the blood-dependent eye is sub-optimal (deficient) there — the regulated eye is protected. But regulation caps the cell's *endogenous* production; it cannot oppose an *external* formate flood. As blood formate climbs into the methanol range, circulating formate alone saturates local insulin into excess in both eyes, and net retinal health collapses to the same insulin-excess floor ($\sim$0.24). The two curves are indistinguishable above $\sim$15 mg/L. Thus the properly regulated retina is healthy and buffered under normal one-carbon fluctuations, and methanol optic toxicity appears as an acute overload that overwhelms that regulation — not as a chronic mis-set of local insulin. (As throughout the eye sections the magnitudes depend on the assumed feedback and window; the robust content is the two-regime behaviour — physiological protection, methanol override.)

Dietary formate and the retina

With the regulated eye in hand we can place dietary formate on the retinal map. The formate-limited retinal cell raises its local insulin modestly with dietary formate (Fig 16), so the distinctive nutritional prediction of the formate theory is ocular, not glycaemic, and remains untested. Reassuringly, dietary eye insulin stays below the retinal optimum — on the beneficial, rising limb of the window, well short of the insulin excess that requires the supra-dietary formate of over-supplementation or methanol. The honest conclusion is that food formate is real but a minor systemic player, and that "eat for your formate" is, if anything, advice for

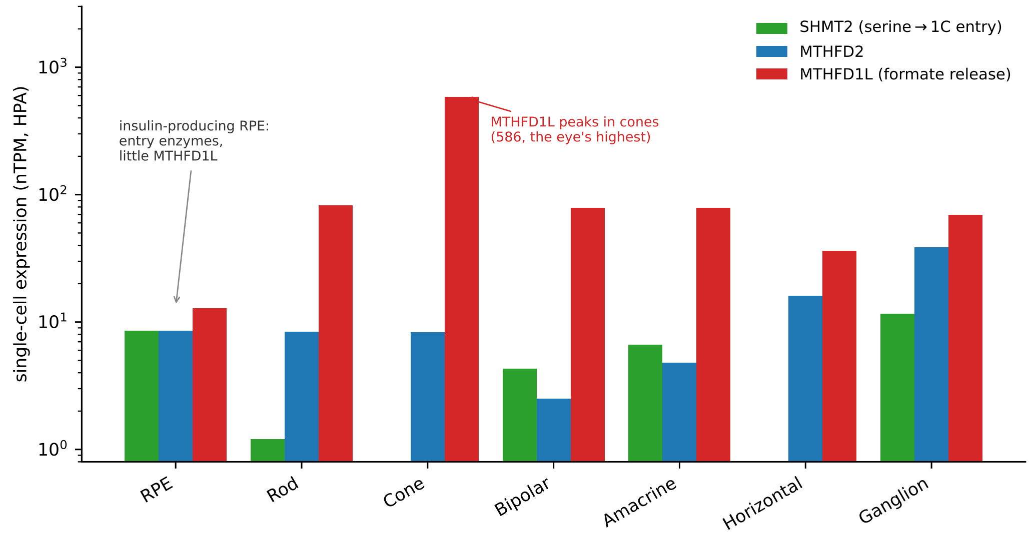

Fig 11. One-carbon/formate enzymes across eye cell types. Single-cell expression (Human Protein Atlas, log scale) of SHMT2 and MTHFD2 (one-carbon entry) and MTHFD1L (formate release). The insulin-producing RPE has the entry enzymes but little MTHFD1L; the formate-releasing enzyme peaks in cone photoreceptors — the outer segments the RPE phagocytoses.

the retina.

Therapeutic implications

The model's map of formate states points to interventions at each level. We consider four: two systemic — nutritional supplementation and metformin — and two ocular — targeting the retina's local formate synthesis, or its insulin synthesis.

Who benefits from formate/serine supplementation?

Start with the systemic question: would a diabetic patient benefit from formate (or its precursor serine) supplementation? We swept the dietary intake in each patient type and read the fasting glucose and the uric-acid side effect (Fig 17). The answer is conditional on β-cell formate status. A *formate-replete* secretion defect does not benefit: the β-cell is already formate-saturated, so extra formate does not raise insulin and fasting glucose is unchanged — it only raises uric acid. A *formate-deficient* diabetic β-cell (low mitochondrial formate: an MTHFD1L variant, antibiotic or microbiome depletion, or mitochondrial dysfunction) is formate-limited and *is* rescued: supplementation raised insulin (+27 %) and lowered fasting glucose (7.0 → 6.5 mM, −8 %). Obesity blunts this benefit (−3 %) because the expanded adipose sink competes for the supplemented formate, and it amplifies the uric-acid side effect (3.5 → 4.7 vs 3.4 → 3.9 mg/dL in the lean). The model therefore predicts a *responder phenotype* — the lean, formate-deficient diabetic — and a non-responder / adverse phenotype — the obese, formate-replete diabetic, in whom supplementation does not help and worsens hyperuricaemia. It also names the stratifying measurements: β-cell/blood formate status (or its causes: MTHFD1L genotype, recent antibiotics) up front, and serum urate for safety.

Metformin: a mitochondrial inhibitor with a dual action

Metformin — the antidiabetic drug of the METTEN trial that seeded the formate–glucose cohort work — is a mitochondrial complex I inhibitor and a PLP-competitive inhibitor of the mitochondrial one-carbon enzyme SHMT2, so it lowers serine → formate flux [35]. We modelled both actions in the diabetic patient: a peripheral, AMPK-like rise in glucose effectiveness (its therapeutic arm) and a reduction in mitochondrial formate production (systemic and β-cell). Across the dose range (Fig 18) fasting glucose fell (−8 % lean,

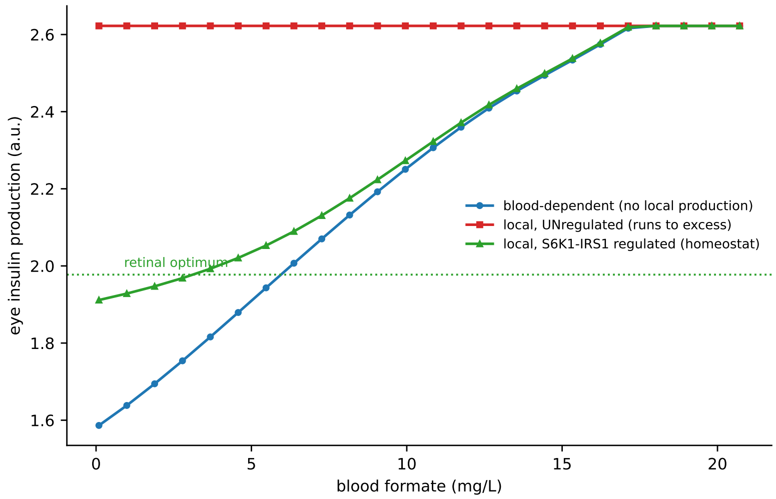

Fig 12. Local co-induced formate: runaway versus regulated homeostat. Predicted eye insulin versus blood formate at fixed glucose. Blood-dependent (blue) rises with circulating formate. The unregulated ATF4 feed-forward (red) runs away to the formate-saturated maximum, in the insulin-excess zone above the retinal optimum. Adding the S6K1 $\rightarrow$ IRS-1 negative feedback (green) makes local production homeostatic: across the physiological range eye insulin is held near the optimum — lifted toward it at low blood formate, where the blood-dependent eye is deficient — and is overridden into excess only by supra-physiological (methanol-range) external formate. The response is monostable throughout (no bistability/hysteresis; `eye_bifurcation.py`).

$-5\,\%$ obese) — yet plasma insulin *fell* too, and blood formate fell by $\sim$24 %. The model thus recovers metformin's defining clinical feature, that it lowers glucose *without* stimulating insulin (indeed it is insulin-sparing), by locating the benefit in the peripheral arm; and it predicts, as an experimentally checkable signature, that metformin *lowers* circulating formate through SHMT2 inhibition — opposite in sign to formate supplementation. A corollary worth flagging is that metformin, by draining mitochondrial formate, could deepen a pre-existing β-cell formate deficiency; its net benefit here nonetheless remains positive because the peripheral arm dominates the fasting glucose.

An intravitreal MTHFD2 inhibitor: a targeted retinal therapy, and a test of the mechanism

The regulated eye gives the model a druggable node. MTHFD2, the NAD^+-dependent mitochondrial methylenetetrahydrofolate dehydrogenase/cyclohydrolase, is the entry enzyme of the pathway that produces the retina's *local* formate (with MTHFD1L releasing it); it is strongly upregulated in tumours but near-absent in most normal adult tissue, and potent, selective, orally available inhibitors have been developed as anticancer agents (e.g. DS18561882 [36]), with MTHFD2 already flagged as an ocular drug target in retinoblastoma [37]. In the model an intravitreal MTHFD2 inhibitor scales down the eye's local mitochondrial formate production (both the endogenous floor and the ATF4-induced term) while leaving systemic glucose, plasma insulin and blood-delivered formate untouched — a local intervention on exactly the feed-forward of the eye-compartment model. We swept the inhibition from 0 to 100% (Fig 19).

The prediction is a precision therapy with a sharp patient-selection rule, and a clean mechanistic control. In a retina carrying *excess* local insulin — the lean secretion-defect overflow with hyperglycaemia, the retinopathy-prone state of the eye-compartment model (eye insulin $\sim$2.46, above the optimum) — lowering local formate walks insulin back down through the optimum, and net retinal health rises from $\sim$0.30 to its peak $\sim$0.42 at $\sim$70–80% inhibition (Fig 19A,B, red): a local, glucose-sparing route to normalising insulin-excess retinopathy, distinct from anti-VEGF. But the same drug is *harmful* to a formate-*deficient* retina (obesity, metformin — already below the optimum): it deepens the deficiency, and net health falls from $\sim$0.42 to $\sim$0.19 (Fig 19B, blue). MTHFD2 inhibition is thus the mirror image of formate/serine supplementation (above): it helps the insulin-excess eye and harms the deficient one, and demands the same

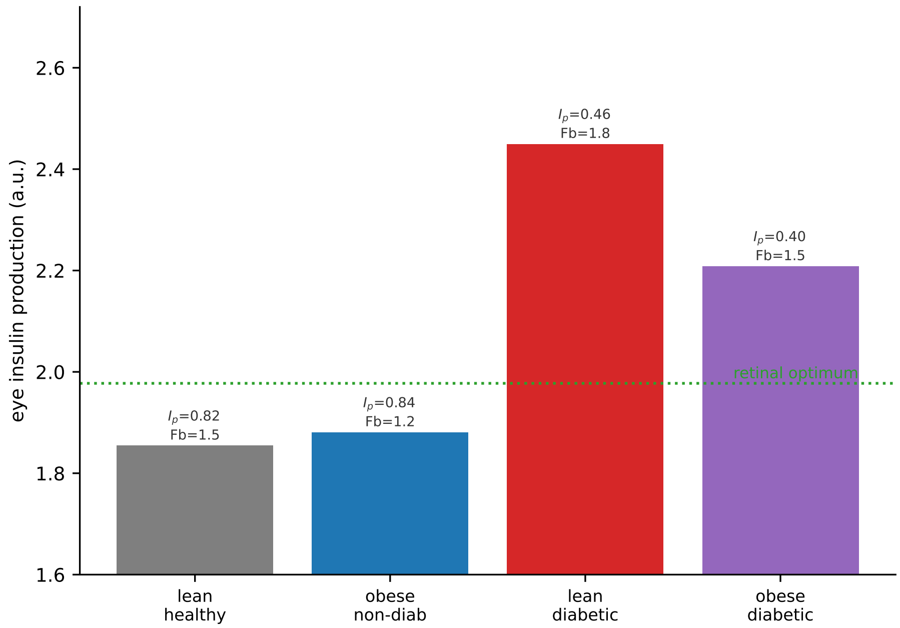

Fig 13. Regulated eye insulin near the optimum, decoupled from plasma insulin. Regulated eye insulin across the four clinical conditions (bars), with each state's plasma insulin (I_p) and blood formate (Fb, mg/L) annotated and the retinal optimum marked. Eye insulin *anti*-correlates with plasma insulin ($r = -0.90$) and tracks blood one-carbon and local glucose: the high-I_p lean-healthy and obese states sit just below the optimum (mild deficiency), while the low-I_p lean secretion-defect diabetic — formate overflow plus hyperglycaemia — is pushed above it into mild excess.

stratification by retinal one-carbon/insulin status. Over-inhibition even of an excess retina eventually overshoots into deficiency, so the effect is dose-limited.

Against methanol optic toxicity the model makes a decisive *negative* prediction (Fig 19C). Because the toxic formate of methanol poisoning is generated *systemically* (hepatic oxidation) and floods the retina across the blood-retinal barrier, it bypasses the retinal MTHFD2 entirely: at intoxication levels eye insulin and net retinal health are *identical* with and without the inhibitor (both at the insulin-excess floor $\sim$0.24), and at physiological formate the inhibitor only *worsens* the retina by inducing local deficiency. An MTHFD2 inhibitor therefore cannot rescue methanol blindness in this model — a result that both cleanly separates endogenous (local, druggable) from exogenous (blood-borne, non-druggable) formate and offers a direct test of the whole eye mechanism: if retinal insulin excess is the driver, targeting local formate synthesis should help the diabetic-overflow retina but not the methanol-flooded one.

An insulin-production inhibitor rescues methanol optic toxicity

The MTHFD2 inhibitor failed against methanol because it acts *upstream* of the formate flood; the natural counter-question is whether acting *downstream*, at the insulin-synthesis node itself, would succeed. In the model insulin production is mTOR-driven translation, so the direct pharmacology is an mTORC1 inhibitor (rapamycin/sirolimus), which reduces β-cell proinsulin biosynthesis [38] and is available for the eye (intravitreal sirolimus has been trialled in retinal disease); secretion-blocking classes (diazoxide on K_{ATP}, somatostatin analogues such as octreotide, the latter with intraocular delivery under study) are an alternative acting on release rather than synthesis. We model the synthesis inhibitor as mtor_mult < 1, capping the mTOR-driven insulin translation downstream of formate and purine sensing, and ask how the regulated eye behaves under a methanol load (Fig 20).

It rescues the toxicity. Untreated, the eye collapses to the insulin-excess health floor ($\sim$0.24) once blood formate exceeds $\sim$50 mg/L; a $\sim$50% intravitreal mTOR inhibitor holds local insulin at the optimum and net retinal health at its *peak* ($\sim$0.42) across the entire methanol range, up to the lethal 500 mg/L (Fig 20A,

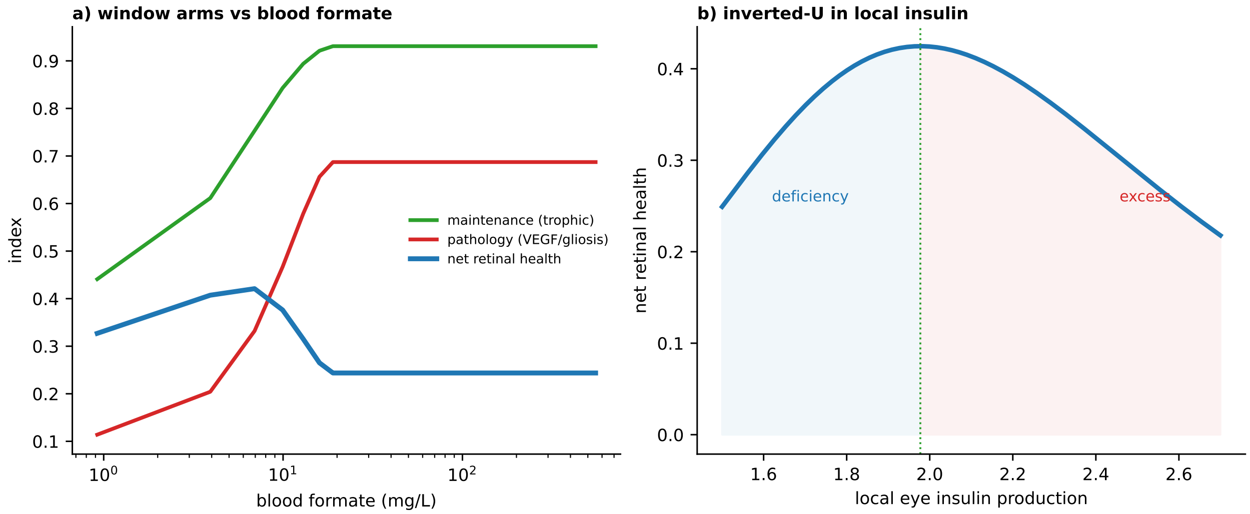

Fig 14. Retinal health is an optimal window in local insulin. Retinal health is scored as a trophic maintenance arm minus a higher-threshold pathology arm (VEGF/gliosis/neovascularization); no complex-IV inhibition is invoked. (a) across blood formate, maintenance, pathology and net health (markers at 5, 50, 500 mg/L); (b) net health is an inverted-U in local eye insulin — both deficiency (low formate) and excess (high formate) harm the retina.

blue). The contrast with the MTHFD2 inhibitor (Fig 20A, dashed — indistinguishable from no drug at intoxication) is the point: because the mTOR inhibitor caps synthesis *downstream* of the purine/formate signal, it clips a blood-borne overload that an upstream, formate-directed drug cannot touch. The dose-response at lethal formate is a therapeutic window (Fig 20B): health recovers to its peak at $\sim$50% inhibition and then declines as over-inhibition overshoots into deficiency. And the margin is wide: because insulin synthesis has a mTOR-independent floor and saturates at the top, a proportional mTOR cap preferentially clips the saturated *excess* while barely moving the near-optimal physiological eye (net health 0.422 $\rightarrow$ 0.417 at physiological formate; Fig 20C) — unlike the MTHFD2 inhibitor, which harmed the physiological retina. The prediction is therefore specific and testable: during methanol intoxication an intravitreal insulin-synthesis (mTOR) inhibitor should preserve retinal function in this insulin-excess model, whereas a local-formate (MTHFD2) inhibitor should not — a discriminating experiment for the insulin-excess mechanism of methanol blindness itself. (As throughout, the magnitudes depend on the assumed window and feedback; the robust content is the directionality — a downstream synthesis block rescues a blood-borne overload that an upstream block cannot — and the acute, local nature of the intervention sidesteps the systemic β-cell liabilities of chronic mTOR inhibition.)

Discussion

A kinetic model shows that the proposed formate $\rightarrow$ purine $\rightarrow$ mTOR $\rightarrow$ insulin chain is quantitatively coherent and, once the glucose–insulin axis is calibrated to islet data, yields non-trivial predictions: a formate dose-response of insulin synthesis, a $\sim$45 % insulin deficit under mitochondrial formate loss, coupled uric-acid changes, and dietary rescue. The sensitivity analysis pinpoints the mTOR purine-sensing Hill coefficient and the mitochondrial formate capacity as the parameters most in need of measurement.

Absolute insulin quantities are now anchored: scaling the granule pool to the measured β-cell insulin content ($\sim$20 pg/cell) gives a stimulated secretion of $\sim$16 pg/cell/h ($\sim$2.8 fmol/cell/h), of the expected order; this single scale does not affect any calibrated ratio. The peripheral alternative, previously untested, is now represented explicitly in the whole-body loop and yields the plasma-insulin discriminator above. Two caveats remain. First, the insulin-secretion and purine-degradation kinetics outside the reused core are still scaled rather than individually fitted, though the sensitivity analysis shows the predictions do not depend on them.

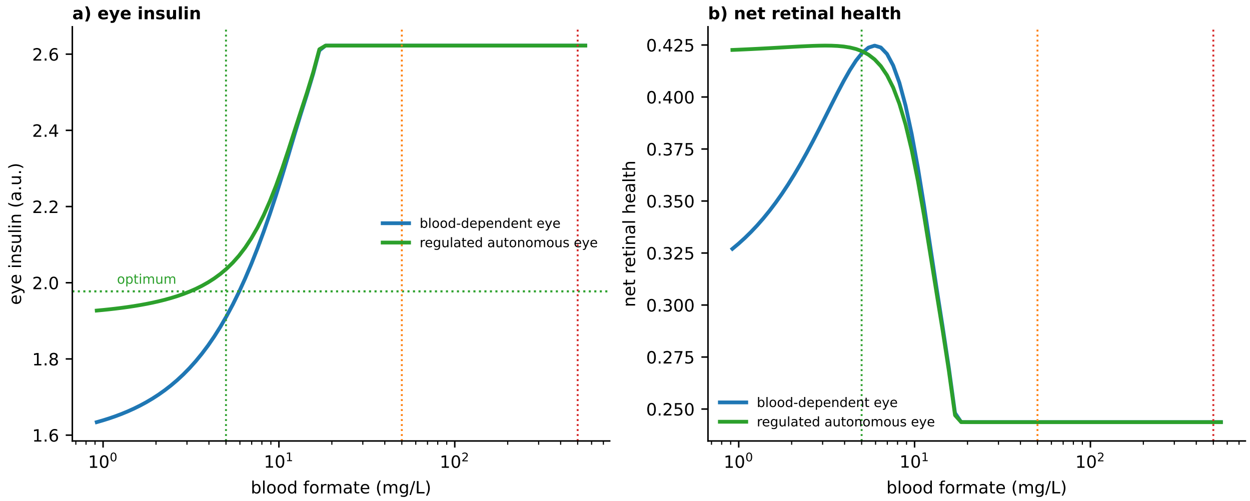

Fig 15. Regulated autonomous eye under methanol (insulin-excess only, no complex-IV inhibition). Blood formate swept to the methanol range (log axis; markers at 5, 50, 500 mg/L). (a) eye insulin — the S6K1-regulated autonomous eye holds near the optimum at physiological formate, then follows the blood-dependent eye up as circulating formate floods in. (b) net retinal health — the regulated eye is at its healthy peak across the physiological range (buffered, unlike the deficient blood-dependent eye) but collapses to the same insulin-excess floor once the methanol flood overrides regulation.

Second, the whole-body layer is a minimal (Bergman-type) representation; a physiological multi-tissue model would refine the quantitative split. The discriminating experiment stands regardless: measure plasma insulin (and, in islets, proinsulin/*INS* translation across a formate gradient, and GSIS in MTHFD1L-deficient islets) during formate supplementation.

Confrontation with existing data. A survey of candidate test datasets finds the mTOR → insulin arm strongly supported (Raptor/TSC2 β-cell models; rapamycin) [17, 38, 39] and folate/one-carbon deficiency impairing β-cell insulin biosynthesis [40], but also two findings in tension with the model that must be addressed. First, in islets, acute knockdown of the mitochondrial one-carbon pathway (SHMT2/MTHFD2) *increases* glucose-stimulated secretion while its ATF4-driven induction raises insulin *content* but lowers secretion [30] — a synthesis-versus-secretion dissociation that this model is built to represent and that argues for measuring content/translation, not only GSIS. Second, a large prospective cohort reports *higher* circulating formate with incident diabetes (opposite the "low-formate" reading of Pietzke et al. [11]); as shown above (Fig 6) the model reproduces this through the secretion-defect overflow route, and the cohort's own data (formate independent of genetic risk and not a risk mediator) place formate as a consequence marker, consistent with that route. Higher serum urate likewise associates with diabetes although Mendelian randomization is null — so the urate arm is best read as a flux marker of purine turnover, not a health claim. Each remains an open test that the paired measurements above would settle.

Materials and methods

This section summarises the models; S1 Appendix gives the complete specification — every state variable, rate law, ODE balance, parameter value with provenance, calibration procedure and the monostability analysis (Fig A in S1 Appendix). Fifteen ODEs (ATP, ADP, AMP, formate, 10-CHO-THF, PRPP, GAR, AICAR, IMP, adenylosuccinate, guanine nucleotides, hypoxanthine/xanthine, urate, pro-insulin, insulin) with mass-action/MM/Hill rate laws; time unit hours, concentrations mM (insulin in arbitrary units). Integrated with an implicit solver (BDF); steady states from a pre-integration followed by a log-space least-squares polish (residuals $< 10^{-13}$ mM/h). The purine/energy core and the PPAT–NUDT5 glue calibration are inherited from Formate-NUDT5 [41]; GSIS parameters were fitted here by least squares to EC_{50} and stimulation-index targets. Insulin quantities are converted to per-cell units by one scale anchoring the

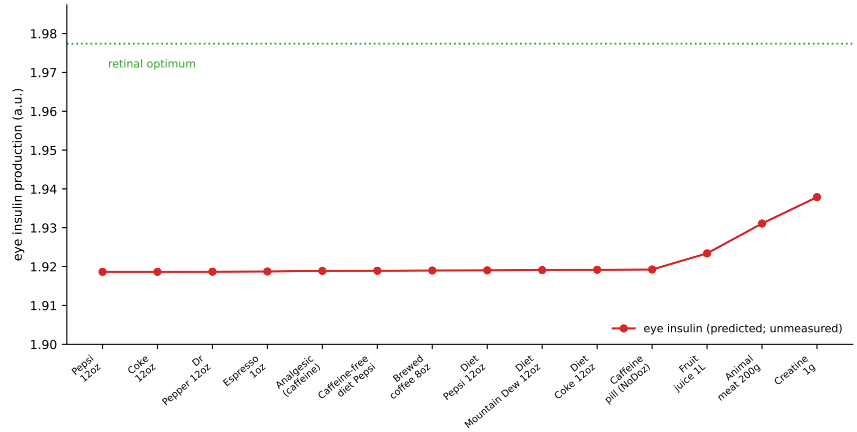

Fig 16. Eye insulin by food (systemic insulin flat). Predicted eye (local retinal) insulin production per food (a.u.); plasma/systemic insulin is flat across all foods — the β-cell is formate-replete, consistent with the measured null effect of creatine and aspartame on fasting insulin — and is not shown. The reference line marks the retinal optimum (peak net retinal health). Dietary eye insulin rises modestly with food formate but stays below the optimum, on the beneficial limb of the window; it is an untested prediction.

granule pool to the measured β-cell insulin content. The whole-body loop (`WholeBody`) embeds the β-cell as the insulin source of a Bergman minimal model (glucose effectiveness S_g, insulin sensitivity S_i, remote insulin action), with per-hour rate constants; the peripheral formate effect is a fractional increase in S_g. For the diabetes-paradox analysis (`FormateDiabetes`) blood formate is an additional dynamic pool fed by dietary intake and endogenous production and drained by an insulin-scaled consumption term and renal clearance, and the secretion defect scales plasma insulin appearance. The obesity analysis (`CoupledModel`) adds a third compartment — an adipocyte metabolic model (`AdipocyteModel`, ten ODEs: the shared energy and one-carbon/purine modules with insulin-dependent glucose uptake and lipid storage, and XOR-mediated urate secretion) — coupled to the β-cell through shared blood glucose, insulin, formate and urate pools; obesity scales adipose mass, XOR and insulin resistance, and the supplementation analysis sweeps the dietary intake with the β-cell held formate-replete or formate-deficient. Metformin is modelled as a rise in glucose effectiveness (peripheral/AMPK) together with a reduction in mitochondrial formate production (systemic and β-cell), representing SHMT2 inhibition. The eye compartment (`EyeModel`) is a retinal insulin cell using the β-cell machinery driven by blood glucose and formate with a low endogenous mitochondrial formate and strictly local insulin (no plasma exchange). Local formate production is an ATF4-style feed-forward on the cell's own mTOR/insulin signal, capped by an mTORC1–S6K1–IRS-1 negative feedback; feed-forward steady states are obtained by integrating to the stable attractor (not root-polished, which can return dynamically unstable fixed points), and the regulated eye was verified monostable by basin tests and a self-consistency/bifurcation analysis (`eye_bifurcation.py`). Retinal health is a trophic-maintenance minus higher-threshold pathology (VEGF/gliosis) inverted-U in local insulin; methanol is modelled purely as this insulin-excess overload, with no complex-IV inhibition. Parameters carry provenance tags [PUB]/[FIT]/[LIT]/[ASM]. The eye analyses (local retinal insulin, the insulin-excess optimal window, methanol overload, and the regulated local formate feed-forward) use `EyeModel`; the dietary meta-analysis maps the per-food formate content to model intake and compares model predictions with pooled literature estimates. The model is implemented as an object-oriented class hierarchy in `model.py`: a `MetabolicCell` base (shared BDF integration, log-space steady state, and rate-law primitives) subclassed by `BetaCellModel`, `AdipocyteModel` and `EyeModel`, with `WholeBody`, `FormateDiabetes`, `CoupledModel` and its gut-microbiome extension `GutUrateCycle` containers; each compartment carries an explicit parameter dataclass. Code, calibrated outputs (`data/*.csv`) and figure generators (`make_figs.py`, `make_body.py`, `make_diet.py`, `make_eye_figs.py`, `make_mthfd2.py`, `make_mtor_inhibitor.py`, `eye_bifurcation.py`, `make_supp_bif.py`) are in this repository.

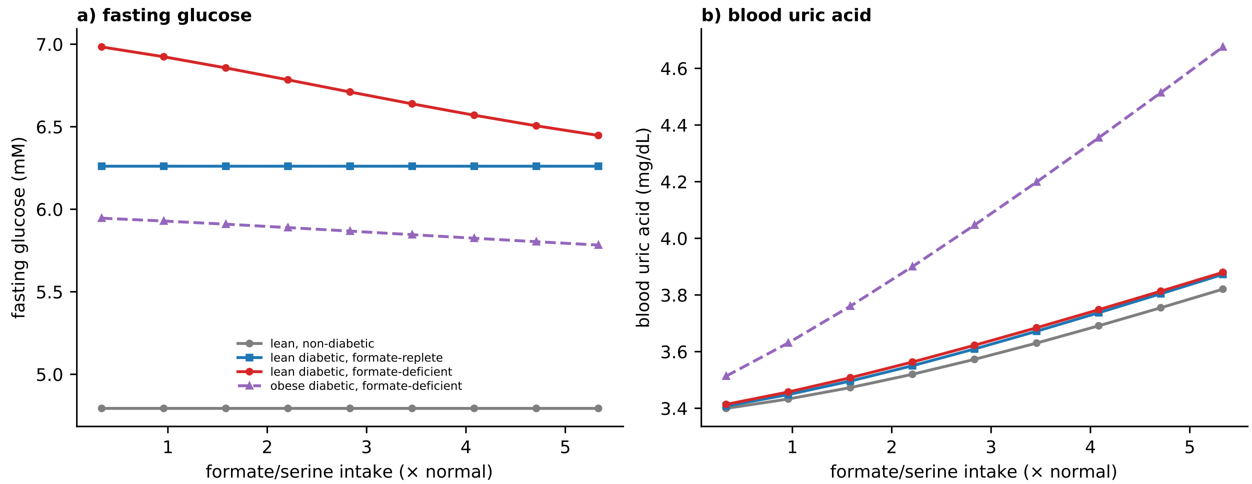

Fig 17. A responder phenotype for formate/serine supplementation. Fasting glucose (a) and blood uric acid (b) versus supplementation dose. Only the formate-deficient diabetic β-cell is rescued (glucose falls); the formate-replete diabetic and the non-diabetic do not respond. Obesity blunts the glucose benefit (adipose sink competition) and worsens the uric-acid side effect.

Supporting information

S1 Appendix. Supplementary methods and full model specification. Complete specification of every state variable, rate law, ODE balance and parameter value with provenance; the calibration procedure; and the monostability analysis of the regulated eye (basin tests and self-consistency/bifurcation analysis, Fig A).

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Data availability

All code, calibrated data outputs and figure-generating scripts underlying the results are openly available in the accompanying project repository: the object-oriented model (`model.py`), the figure generators (`make_*.py`, `eye_bifurcation.py`) and the calibrated outputs (`data/*.csv`). Repository: `https://github.com/av2atgh/formate` (archived at Zenodo, DOI to be assigned on submission).

Competing interests

AV is a paid employee of Nodes & Links Ltd. This does not alter the author's adherence to PLOS Computational Biology policies on sharing data and materials.

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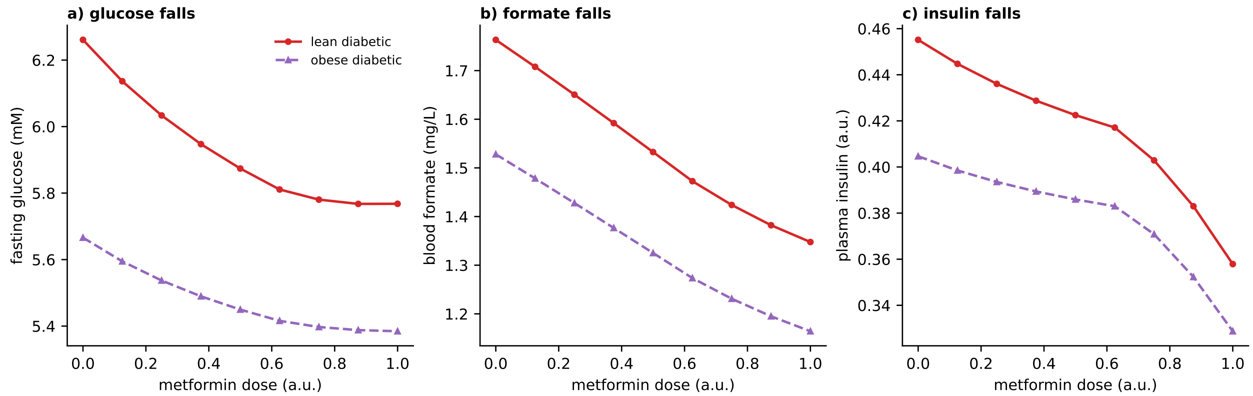

Fig 18. Metformin lowers glucose peripherally while lowering formate and insulin. Dose response in lean and obese diabetics. (a) fasting glucose falls (peripheral/AMPK arm); (b) blood formate falls (SHMT2 inhibition of mitochondrial formate); (c) plasma insulin falls — metformin is insulin-sparing, not a secretagogue.

Table 1. Key numbers (this run).

Quantity	Value
ATP/ADP, 1→10 mM glucose [14]	2.4→11.6 (target 2.4/11.6)
IMP / S-AMP, low → high glucose [16]	×0.28 / ×3.4 (target 0.23/3.4)
Raptor-KO insulin content [17]	50% of WT (target 50%)
GSIS EC_{50} / stimulation index	8.0 mM / 5.5 (targets 8.0 / 5.5)
Insulin synthesis, formate 0 → 2 mM (11 mM glucose)	+66 %
Insulin synthesis loss, mito-formate 100% → 0%	∼45 %
mTOR, mito-formate 100% → 0%	0.35 → 0.03
Urate secretion, mito-formate 100% → 0%	0.10 → 0.02 mM/h
Insulin content / secretion (absolute)	20 pg/cell / 16 pg/cell/h
Central formate: Δglucose AUC / Δinsulin AUC	−21 % / +20 %
Peripheral formate: Δglucose AUC / Δinsulin AUC	−14 % / −2 %
Blood formate: obesity / diabetes (vs healthy)	−14 % / +21 %
Supplementation: Δglucose, deficient / replete diabetic	−8 % / 0 %
Metformin: Δglucose / Δblood formate	−8 % / −24 %
Eye insulin: corr. with blood formate / plasma insulin	+0.89 / −0.90
Regulated eye: retinal health, physiological / methanol range	0.42 (peak) / 0.24 (floor)
Diet: corr(food formate, T2D log-RR)	−0.02 (no relationship)

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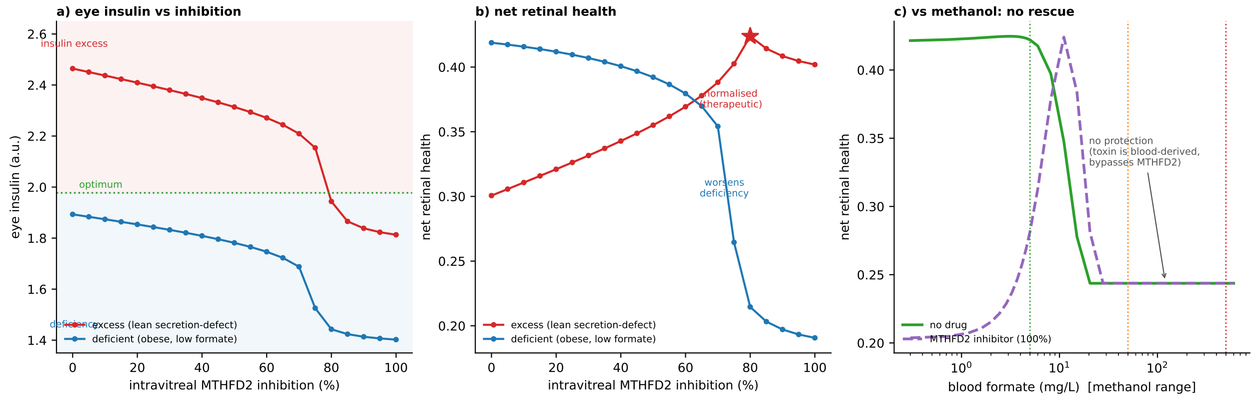

Fig 19. An intravitreal MTHFD2 inhibitor normalises insulin-excess retinopathy but cannot rescue methanol toxicity. Retinal MTHFD2 inhibition scales down local mitochondrial formate only (blood formate still crosses the barrier). (a) Eye insulin falls with inhibition; the excess (lean secretion-defect) retina (red) descends through the optimum, the deficient (obese/low-formate) retina (blue) into deficiency. (b) Net retinal health: partial inhibition (∼70–80%, star) *normalises* the excess retina (therapeutic), while the same dose *worsens* the deficient retina. (c) Against a methanol formate flood, the inhibitor gives no protection — health with (dashed) and without (solid) the drug coincide at the insulin-excess floor once blood formate overrides local synthesis — because the toxin is blood-derived and bypasses MTHFD2.

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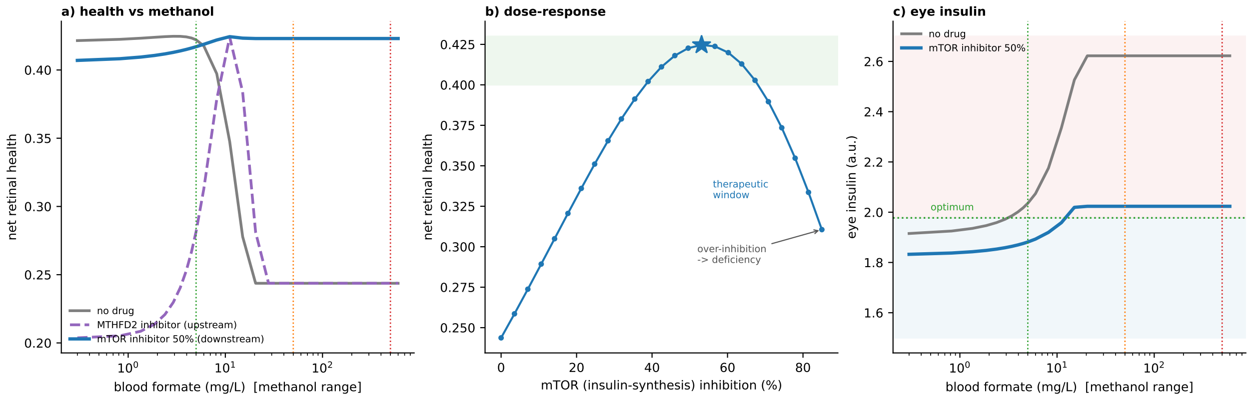

Fig 20. An intravitreal insulin-production (mTOR) inhibitor rescues methanol optic toxicity, where the MTHFD2 inhibitor could not. Insulin synthesis is mTOR-driven, so an mTORC1 inhibitor (mtor_mult < 1) caps it downstream of the formate flood. (a) Net retinal health vs blood formate: no drug (grey) and the upstream MTHFD2 inhibitor (dashed) both collapse to the insulin-excess floor at methanol levels, whereas a 50% mTOR inhibitor (blue) holds health at the optimum across the whole range. (b) Dose-response at lethal methanol (500 mg/L): a therapeutic window peaking near 50% inhibition, over-inhibition overshooting into deficiency. (c) Eye insulin: the mTOR inhibitor clips the methanol excess (down to the optimum) while sparing the near-optimal physiological baseline.

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